

Cognition-aligned frequency filtering for sentence embeddings[☆]Chenrui Mao^{a,*,} Kai Shuang^{a,*}, Jinyu Guo^{c,*}, Bing Qian^b, Yu Yang^a, Haoqing Li^a^a Beijing University of Posts and Telecommunications, Beijing, 100876, China^b China Telecom Corporation Limited Beijing Research Institute, Beijing, 102209, China^c School of Information and Software Engineering, University of Electronic Science and Technology of China, Chengdu, 610054, China

ARTICLE INFO

Keywords:

Semantic textual similarity

Sentence embedding

Contrastive learning

Embedding frequency filtering

ABSTRACT

Learning better sentence embeddings that capture precise semantic plays an important role in Natural Language Processing (NLP). The Sentence Textual Similarity (STS) of embeddings reflects their semantic precision, as this task requires a direct comparison of semantic meanings in vector space. Thus, we focus on improving the ability of sentence embeddings to capture semantic similarity. From the perspective of human cognition, we identify a critical cognitive gap in frequency-domain semantic representation: while semantic information is distributed across all frequency components of embeddings, the human selective attention mechanism suggests that only specific frequency bands are utilized for semantic processing. This frequency-domain cognitive gap leads to semantic redundancy in machine-learned embeddings, which is particularly detrimental for tasks requiring redundancy-resistant representations. To bridge this gap, we propose a simple Cognition-Aligned Frequency Filtering (CAFF) method for unsupervised embedding training on 10⁶ sentences from Wikipedia. CAFF introduces a self-adaptive Frequency Filtering Unit (FFU) to modulate the frequency components of embedding. The FFU functions as a filtering mechanism that suppresses irrelevant components in embeddings to mitigate semantic redundancy. Extensive evaluations with SentEval show that our embeddings improve over the initial encoder by 2.33% on the STS task, achieving state-of-the-art performance. Additionally, our results demonstrate improved performance on both transfer and retrieval tasks.

1. Introduction

Machine learning aims to mimic human cognition. Serving as the entry point to Natural Language Processing (NLP), universal sentence embeddings remain a fundamental challenge in NLP (Conneau et al., 2017; Gao et al., 2021; Kiros et al., 2015; Ou & Xu, 2024; Reimers & Gurevych, 2019; Xu et al., 2023). High-quality embeddings should preserve rich semantic information while demonstrating robust performance across diverse downstream tasks without requiring task-specific fine-tuning (Chuang et al., 2022). Among these downstream tasks, Semantic Textual Similarity (STS) task provides the most direct evaluation of embedding quality through pairwise semantic comparisons between embeddings. Although pre-trained embeddings achieve suboptimal performance in many downstream tasks, they perform poorly on the STS task prior to fine-tuning, indicating inherent limitations in their semantic representation capabilities.

[☆] This work was supported by National Key R&D Program of China (Grant No. 2024YFF0618700) and Postdoctoral Fellowship Program (Grade C) of China Postdoctoral Science Foundation (Grant No. GZC20251053)

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<https://doi.org/10.1016/j.ipm.2025.104415>

Received 30 April 2025; Received in revised form 4 September 2025; Accepted 19 September 2025

Available online 27 September 2025

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This limitation may stem from a fundamental cognitive mismatch between machine-learned representations and human cognition in semantic processing. While semantic similarity aligns with feature representations that mirror human cognition (Miller & Charles, 1991; Tversky, 1977), existing representation learning objectives often neglect the critical role of selective attention mechanisms (Desimone & Duncan, 1995), which have been extensively studied in cognitive psychology and neuroscience. Classic models of selective attention, such as Broadbent's filter theory (Broadbent, 1958) and Treisman's attenuation theory (Treisman, 1960), posit that human perception is constrained by limited processing capacity, necessitating the prioritization of task-relevant sensory information while attenuating less pertinent signals. Neural mechanisms underlying selective attention (Corbetta & Shulman, 2002; Desimone & Duncan, 1995) indicate that the human brain dynamically prioritizes features such as lower or specific frequencies in both auditory (Cherry, 1953) and visual domains (Corbetta & Shulman, 2002), while suppressing others as noise or redundancy. In contrast, machine-learned representations do not intrinsically possess this selective attention mechanism, often retaining all frequencies regardless of their relevance. This difference underpins a fundamental cognitive gap which we term **frequency redundancy** between machine and human semantic processing. As documented in the Computer Vision (CV) domain Fuoli et al. (2021), Wang et al. (2020), Zhou et al. (2022), humans tend to perceive low-frequency components, while models often exploit high-frequency components that are imperceptible to humans. To bridge this gap, recent approaches (Guo et al., 2023; Lin et al., 2023; Lu et al., 2023) have introduced frequency attention and filtering networks, aligning model representations more closely with human perception and thereby boosting downstream task performance.

From this perspective, we hypothesize that an analogous frequency redundancy may hinder NLP representations, which are also based on neural networks similar to those in CV. The performance bottleneck of pre-trained embeddings in tasks such as semantic textual similarity (STS) has already indicated the limitations imposed by this gap. Different from CV, in the context of our work, "frequencies" (or frequency bases) refer to the components obtained by applying a frequency transform to the representation vectors (i.e., word or sentence embeddings). The intensity of each frequency refers to the modulus of the corresponding coefficient in the transformed domain. Specifically, the frequency bases decompose an embedding into a set of oscillatory patterns that capture information at different granularities: lower frequency components typically encode global or coarse-grained semantic information, while higher frequency components capture localized or fine-grained variations (Bracewell, 1966; Williamson, 1999). Importantly, these frequency bases can be interpreted as latent semantic features that correspond to different levels of abstraction in human cognition. While current contrastive learning frameworks for sentence embeddings and their enhanced variants (Chen et al., 2023; Jeong et al., 2024; Zhu et al., 2024) strive to align the training objective with the human cognition's feature-matching process (Tversky, 1977), they nonetheless continue to operate on the full frequencies of embeddings, rather than selectively emphasizing certain frequencies as human cognition would. Thus, the persistence of unselected frequencies (i.e., frequency bases that are irrelevant according to the learned mask and suppressed during filtering) in machine-generated embeddings further manifests as **semantic redundancy**. This phenomenon is empirically supported by our findings on the STSB dataset (Cer et al., 2017), where suppressing random frequency components unexpectedly improves semantic similarity scores (from 74.1 to 89.4 on average cosine similarity, as shown in Fig. 1 with an example), indicating that not all spectral information is necessary – or even beneficial – for human-like semantic comparison. These results suggest that the quality of embeddings may benefit from selective frequency modulation, closely mirroring the selective attention mechanism of human cognition. Although our experiments focus on English, the proposed frequency-based analysis is theoretically language-independent, as selective attention and frequency prioritization are universal aspects of human cognition. We consider that, although different languages may have distinct frequency distributions, the mechanisms for selecting and prioritizing frequencies operate in a fundamentally similar way across languages. Therefore, frequency redundancy is expected to generalize beyond English, which here serves as a representative case.

Based on the above discussion, we propose a simple yet effective **Cognition-Aligned Frequency Filtering (CAFF)** method for sentence embeddings to bridge this cognitive gap. Central to CAFF is a **Frequency Filtering Unit (FFU)** that adaptively generates frequency masks – via linear and convolutional operations – emulating the human brain's ability to amplify informative features while attenuating noise (Desimone & Duncan, 1995). The contrastive learning objective encourages embeddings of similar sentences to have similar feature compositions, including their spectral structures, which is consistent with human feature-matching processes (Tversky, 1977). Building on this foundation, we assume that frequency components with high intensity in the embedding are those most likely to be selected by human cognition for semantic discrimination, while low-intensity frequency components are more likely to be regarded as redundant or irrelevant. Thus, we adopt the contrastively pre-trained embeddings as our initial embeddings. Based on these embeddings, the FFU applies self-adaptive masks to selectively enhance selected frequencies (i.e., high-intensity frequencies), while suppressing non-selected frequencies (i.e., low-intensity frequencies) that are typically irrelevant in human cognitive processing. Through spectral modulation rather than arbitrary truncation, CAFF preserves semantic integrity while eliminating semantic redundancy. Furthermore, a subsequent **Feed-Forward Network (FFD)** is introduced to capture inter-frequency relationships for further refinement. Through a hybrid training method that combines contrastive learning with frequency-aware refinement. This enables CAFF to extract, leverage, and retain only those frequency components most relevant for semantic matching, thereby alleviating the aforementioned semantic redundancy issue and leading to higher-quality embeddings.

To verify the effectiveness of our approach, we evaluate the quality of the sentence embeddings using the unified evaluation standard SentEval (Conneau & Kiela, 2018). We demonstrate improvements on STS tasks, transfer tasks, and retrieval tasks through our proposed frequency filtering method. Our trained embeddings yield significant improvements over the initial encoder, outperforming it by 2.33% in STS tasks and reaching state-of-the-art (SOTA) performance.

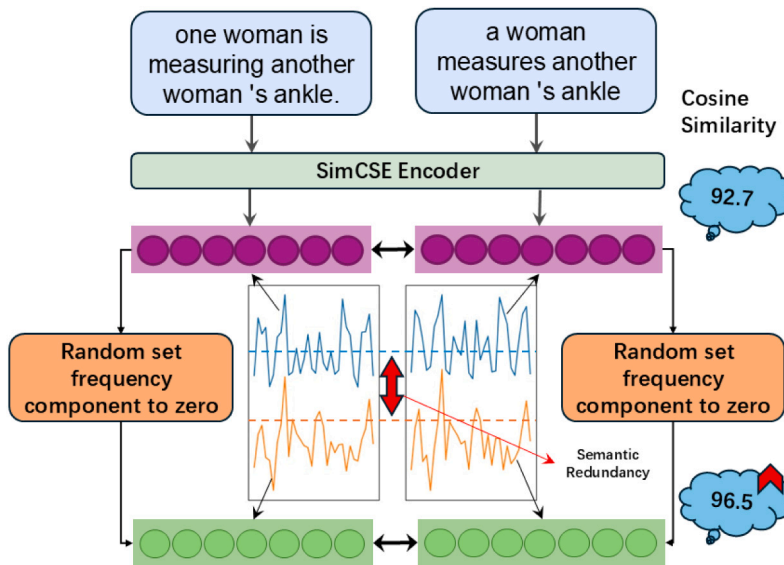


Fig. 1. Illustration of semantic redundancy: When random frequencies are filtered out, the cosine similarity between two sentences increases.

1.1. Research objective and contributions

The objective of our research is to perform a filter-like operation to remove semantic redundancy caused by the cognitive mismatch between machine-learned representations and the frequency-based selective attention previously discussed. This method aims to filter out redundant frequency components in embeddings while preserving semantic integrity. We focus on improving performance on STS tasks as an example, since these tasks require direct semantic comparison. Also, we demonstrate improvements on transfer and retrieval tasks. Our key research objective is to bridge the cognitive gap by incorporating frequency-aware processing, moving beyond traditional full-frequency utilization. Thus, we aim to construct an adaptive frequency filtering unit that modulates embedding frequencies using self-adaptive masks, amplifying task-critical frequencies and suppressing redundant ones. Our main contributions are summarized as follows:

1. We are pioneers in investigating human cognitive principles for potential enhancement of pre-trained embeddings within the NLP field.
2. We design a novel CAFF training method that addresses the cognitive gap in semantic redundancy through a cognition-inspired FFU.
3. Empirical validation shows that CAFF greatly improves the initial encoder on STS tasks, achieving state-of-the-art (SOTA) performance. Additionally, CAFF improves results on transfer and retrieval tasks.

2. Related works

2.1. Embeddings with contrastive learning

Sentence representation learning remains a crucial topic in NLP applications, as it directly supports tasks such as semantic textual similarity (STS), information retrieval, and question answering (Cer et al., 2017; Khashabi et al., 2020). Early studies trained sentence representations based on Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014), which provided vector representations by aggregating word embeddings. With the development of pre-trained language models such as BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), and XLNet (Yang et al., 2020), sentence representations achieved significant improvements. However, despite these advances, pre-trained models still underperform on STS benchmarks, as their embeddings are not explicitly optimized for semantic similarity. To address this limitation, Sentence-BERT (SBERT) (Reimers & Gurevych, 2019) introduced a Siamese network architecture that fine-tunes BERT with sentence-pair supervision, yielding embeddings well-suited for similarity tasks. Following this direction, SimCSE (Gao et al., 2021) proposed contrastive learning to optimize sentence embeddings. Its unsupervised version employs dropout as minimal data augmentation, generating positive pairs by encoding the same sentence twice with different dropout masks, while in-batch negatives provide contrastive signals. The supervised version further leverages Natural Language Inference (NLI) datasets (Bowman et al., 2015; Williams et al., 2018), treating entailment pairs as positives and contradiction pairs as hard negatives, which significantly enhances STS performance. Building on this framework, subsequent research has focused on improving unsupervised sentence embedding learning. Approaches such as ConSERT (Yan et al., 2021), ESIMCSE (Wu et al., 2022), and SSCL (Chen et al., 2023) introduce richer data augmentation strategies, training mechanisms,

and discrete sentence augmentation mechanisms to enhance contrastive learning. Other studies, such as PromptBERT (Jiang et al., 2022) and DiffCSE (Chuang et al., 2022), explore prompt-based encoding or difference-based loss function to obtain more robust semantic representations. More recent work has shifted towards integrating external knowledge (Zhu et al., 2024), inter-layer structural information (Klein & Nabi, 2023), knowledge from large language models (LLMs) (Ou & Xu, 2024) or angle-based loss function (Jeong et al., 2024). Despite these promising advances, existing approaches still neglect the critical cognitive gap between machine learning patterns and human semantic understanding. This underexplored frontier suggests a promising direction for sentence embeddings.

2.2. Frequency domain learning

Frequency domain learning has emerged as a powerful paradigm across multiple deep learning domains, offering advantages in efficiency and interpretable feature representation. Early work (Xu et al., 2020) showed the effectiveness of spectral filters for instance segmentation and object detection. Later, DFF (Lin et al., 2023) introduced adaptive frequency filtering with gradient reversal layers, achieving strong results in Re-identification (ReID) and image classification. These developments build on foundational insights from biological vision systems (Campbell & Robson, 1968). More recent works confirm the benefits of spectral filtering. Frequency-aware transformers improve image compression and super-resolution by decomposing spectral components with low cost (Dai et al., 2024; Li et al., 2024). Hybrid spectral-spatial models also show that FFT operators capture global structure while keeping local detail (Arshad et al., 2025). These studies suggest that frequency priors are widely useful beyond recognition tasks.

In NLP, early studies (Almarwani et al., 2019; Kayal & Tsatsaronis, 2019) explored spectral decomposition for embeddings. FNet (Lee-Thorp et al., 2022) replaced self-attention with Fourier transforms, training much faster than BERT but with lower accuracy, showing its role as an efficiency method rather than a full spectral approach. More recent work makes frequency signals central. Hyena uses FFT-based long convolutions as an alternative to attention, reaching sub-quadratic complexity in language modeling (Poli et al., 2023). FFTNet treats token mixing as learnable spectral filtering with strong scalability (Fein-Ashley et al., 2025). For positional encodings, Fourier Position Embedding (FoPE) strengthens the spectral properties of RoPE and improves length generalization (Hua et al., 2024). Other studies use spectral analysis to detect embedding collapse (Si et al., 2024), to evaluate NLG quality (Yang et al., 2023), or to design filters for better domain generalization (Liu et al., 2025).

Overall, these advances show that frequency-domain priors are becoming actionable in both CV and NLP. Our work builds on this trend by proposing a frequency-aware embedding optimization method with cognitive motivation.

3. Methodology

In this section, we first introduce and analyze the impact of the traditional contrastive learning framework in the frequency domain, and then describe the proposed CAFF method for frequency filtering to obtain higher-quality embeddings.

3.1. Contrastive learning in frequency domain

In the traditional contrastive learning framework, let $\mathcal{D}$ denote the training dataset that consists of sentences from the corpus, i.e., $\mathcal{D} = \{s_1, s_2, \dots, s_n\}$, where s_i denotes an embedding generated by the encoder of a certain sentence. The unsupervised contrastive learning framework treats $s_i^+ = s_i$ and applies different dropout masks to construct positive pairs:

$$x_i = m_1(s_i), \quad (1)$$

$$x_i^+ = m_2(s_i^+), \quad (2)$$

where m_1 and m_2 denote different dropout masks. The loss function is defined as

$$L = -\log \frac{e^{\text{sim}(x_i, x_i^+)/\tau}}{\sum_{j=1}^n e^{\text{sim}(x_i, x_j^+)/\tau}}, \quad (3)$$

where τ is a temperature parameter, and

$$\text{sim}(x_1, x_2) = \frac{x_1 x_2^T}{\|x_1\| \|x_2\|}, \quad (4)$$

denotes the cosine similarity.

Through this training process, the contrastive learning framework pulls semantically close neighbors together and pushes apart non-neighbors (Hadsell et al., 2006). In the frequency domain, the semantic similarity between two embeddings x_1 and x_2 with respect to the frequency transformation matrix F is given by

$$s_f = \frac{(F x_1)(F x_2)^T}{\|F x_1\| \|F x_2\|} = \frac{F x_1 x_2^T F^T}{\|F\|^2 \|x_1\| \|x_2\|}. \quad (5)$$

Note that the similarity function is defined in (4), and thus the semantic similarity in the frequency domain is given by

$$s_f = \frac{1}{\|F\|^2} F \text{sim}(x_1, x_2) F^T, \quad (6)$$

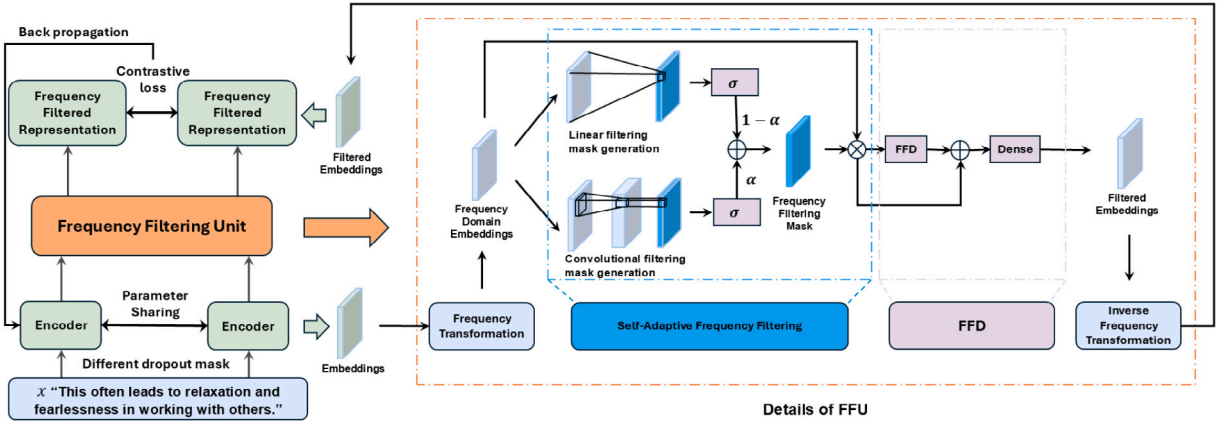


Fig. 2. The training process of our proposed CAFF framework. The FFU filters out low-intensity frequencies as semantic redundancy.

This formula shows that the embedding similarity in the frequency domain is the semantic similarity in the original domain scaled by a fixed constant F . Thus, contrastive learning not only brings closer together the embeddings of two similar sentences in the time domain, but also in the frequency domain. All the properties obtained from contrastive learning in the time domain also hold in the frequency domain. Hence, frequency information is already optimized during contrastive learning. This process is consistent with findings in Tversky (1977), indicating that embeddings trained by the contrastive learning framework can serve as our initial cognition-aligned embeddings. Nevertheless, these embeddings still require further filtering to reduce semantic redundancy. Therefore, in the next step, we build upon these embeddings to achieve more refined representations.

3.2. Cognition-aligned frequency filtering

Through the analysis in Section 3.1, we consider the low-intensity components learned by the traditional contrastive learning framework as concrete representations of semantic redundancy. Accordingly, CAFF regards the high-intensity frequency components as “selected frequencies” in line with human selective attention. Our approach is to enhance the high-intensity frequencies and suppress the low-intensity frequencies. CAFF employs a self-adaptive Frequency Filtering Unit (FFU) to achieve this goal. The training process of CAFF with the FFU is illustrated in Fig. 2.

In CAFF, we use the same data augmentation steps (see Eqs. (1) and (2)) as employed in SimCSE and follow the same procedure to obtain the positive pair x_i and x_i^+ . We then pass the embeddings through the FFU, which consists of a frequency transformation, self-adaptive frequency filtering, a Feed-Forward layer (FFD), and an inverse frequency transformation to obtain the frequency-filtered embeddings:

$$z = F^i (FFD (M(F(x)))) , \quad (7)$$

where F denotes the frequency transformation, F^i denotes the inverse frequency transformation, FFD is the feed-forward layer, and M denotes the filtering mask for self-adaptive frequency filtering. Finally, we use the contrastive loss function to calculate the loss for the frequency-filtered embeddings. The goal of CAFF is to use the FFU to obtain frequency-filtered embeddings for contrastive learning, thereby encouraging the embeddings to focus more on certain frequencies and thus reduce semantic redundancy. Next, we introduce our key component, the FFU, in detail through a step-by-step explanation.

3.2.1. Frequency transformation

Frequency transformation converts embedding vectors into a frequency basis. We treat each embedding x_i as a discrete one-dimensional, non-periodic vector, i.e., $x_i(n) = [x_i(1), x_i(2), \dots, x_i(N)] \in \mathbb{R}^N$, where N denotes the dimensionality of the vector. This vector can be transformed into a discrete one-dimensional frequency domain vector of the same dimension, $f_i(u) = [f_i(1), f_i(2), \dots, f_i(N)]$, via the following transformations:

- **Discrete Fourier Transform (DFT):** DFT uses complex exponentials (sine and cosine functions in complex form) as basis functions. The transformation is defined as

$$f_i(u) = \sum_{n=0}^{N-1} x_i(n) e^{-j \frac{2\pi}{N} un} \quad (u = 0, 1, 2, \dots, N-1), \quad (8)$$

where the result is a complex number. Since complex numbers can be challenging for machine learning models to process, we treat the real and imaginary parts separately.

- **Discrete Cosine Transform (DCT):** DCT uses only the cosine function as the basis, and the transformation is defined as

$$f_i(u) = C(u) \sum_{n=0}^{N-1} x_i(n) \cos\left(\frac{(2n+1)u\pi}{2N}\right) \quad (u = 0, 1, 2, \dots, N-1), \quad (9)$$

where

$$C(u) = \begin{cases} \sqrt{\frac{1}{N}} & u = 0 \\ \sqrt{\frac{2}{N}} & \text{otherwise} \end{cases} \quad (10)$$

and the result is a real number.

- **Discrete Hartley Transform (DHT):** DHT uses the sum of sine and cosine functions as the basis, and the transformation is given by

$$f_i(u) = \sum_{n=0}^{N-1} x_i(n) \left[\sin\left(\frac{2\pi}{N}un\right) + \cos\left(\frac{2\pi}{N}un\right) \right] \quad (u = 0, 1, \dots, N-1), \quad (11)$$

where the result is a real number.

Through the frequency transformation, we map the vector to the frequency domain. In this process, we use the Fast Fourier Transform (FFT) algorithm, which imposes certain requirements on the input dimensionality. Therefore, we apply an MLP layer to transform the embedding to a specific dimension before the frequency transformation:

$$x'_i(n) = \text{MLP}(x_i(n)), \quad f_i(u) = F(x'_i(n)). \quad (12)$$

Here, the MLP layer acts as a mapping from $\mathbb{R}^{d_m} \rightarrow \mathbb{R}^{d_k}$, where d_m denotes the dimension of the original embedding and d_k is a hyperparameter chosen to be compatible with the FFT algorithm. The three aforementioned transformations (DFT, DCT, DHT) represent different levels of frequency decomposition, with DFT providing the most comprehensive decomposition.

3.2.2. Self-adaptive frequency filtering

After obtaining the frequency domain representation, we design a self-adaptive frequency filter to reduce semantic redundancy, which specifically resides in the low-intensity frequency components of sentence embeddings. These low-intensity components may provide little contribution to semantic discrimination, representing irrelevant or noisy information that leads to a cognitive gap between machine-learned representations and human semantic processing. To address this, our filter aims to enhance discriminative “selected frequencies” while suppressing redundant ones. A content-adaptive mask is jointly generated by a convolutional layer, which identifies localized frequency patterns, and a linear layer, which captures global frequency relationships. The outputs of these two layers are weighted and summed to form the final mask, which actively attenuates redundant frequencies and selectively amplifies components critical to semantics. This selective enhancement inherently removes semantic redundancy in the full spectrum by directing model capacity towards informative, task-relevant features rather than uniformly processing all components. The mask generation process is illustrated in Fig. 2.

This frequency filtering mask is constructed based on the frequency vectors of the embeddings, and is therefore referred to as self-adaptive frequency filtering. We generate two distinct filter masks using a convolutional layer and a linear layer, respectively, and then compute their linear combination to obtain the final filter mask. The convolutional filter effectively captures local features in the frequency domain, aiming to enhance high-intensity frequencies while suppressing low-intensity frequencies, thereby simulating frequency domain selectivity and reducing semantic redundancy. Specifically, we construct the convolutional filter using m convolution kernels of size k , followed by a single convolution kernel of size 1. The sigmoid activation function σ is applied after convolution to compute the convolution mask:

$$M_{\text{conv}i}(u) = \sigma(\text{Conv}_{1 \times 1}(\text{Conv}_{9 \times 9}(f_i(u)))). \quad (13)$$

Since the convolutional mask may overemphasize local information, we utilize a linear filter to compensate for the global context. A linear layer with a sigmoid activation function σ is used to generate the linear mask:

$$M_{\text{linear}i}(u) = \sigma(\text{Linear}(f_i(u))). \quad (14)$$

Finally, the two masks are linearly combined to obtain the final frequency filtering mask:

$$M_i(u) = \alpha M_{\text{conv}i}(u) + (1 - \alpha) M_{\text{linear}i}(u), \quad (15)$$

where α is a hyperparameter. This mask is applied to the frequency vectors for frequency filtering:

$$f'_i(u) = M_i(u) \otimes f_i(u). \quad (16)$$

Table 1

Overview of datasets and tasks used in our experiments.

Task	Type	#Train	#Test	Reference
<i>STS Tasks</i>				
STS12	Semantic textual similarity	N/A	3.1k	Agirre et al. (2012)
STS13	Semantic textual similarity	N/A	1.5k	Agirre et al. (2013)
STS14	Semantic textual similarity	N/A	3.7k	Agirre et al. (2014)
STS15	Semantic textual similarity	N/A	8.5k	Agirre et al. (2015)
STS16	Semantic textual similarity	N/A	9.2k	Agirre et al. (2016)
STS-B	Semantic textual similarity	5.7k	1.4k	Cer et al. (2017)
SICK-R	Semantic textual similarity	4.5k	4.9k	Marelli et al. (2014)
<i>Transfer Tasks</i>				
MR	Movie review	11k	11k	Pang and Lee (2005)
CR	Product review	4k	4k	Hu and Liu (2004)
SUBJ	Subjectivity status	10k	10k	Pang and Lee (2004)
MPQA	Opinion-polarity	11k	11k	Wiebe et al. (2005)
SST2	Binary sentiment analysis	67k	1.8k	Socher et al. (2013)
TREC	Question-type classification	6k	0.5k	Voorhees and Tice (2000)
MRPC	Paraphrase detection	4.1k	1.7k	Dolan and Brockett (2005)

3.2.3. Feed-forward module, inverse frequency transformation and loss function

After filtering, we first pass the filtered embeddings through the FFD network before applying the inverse frequency transformation:

$$f_i''(u) = \text{FFD}(f_i'(u)). \quad (17)$$

Since the inverse frequency transformation will be performed after this layer, we remove the batch normalization operation in the FFD network. The FFD network adds a non-linear projection to the filtered embeddings, enabling the capture of nonlinear cross-frequency interactions and further refining the embeddings. After the FFD layer, we transform the embeddings back to the time domain via the inverse frequency transformation:

$$z_i(n) = F^i f_i''(u), \quad (18)$$

where F^i denotes the inverse frequency transformation matrix. Finally, we use the contrastive learning loss function for training:

$$L_f = -\log \frac{e^{\text{sim}(z_i(n), z_i(n)^+)/\tau}}{\sum_{j=1}^n e^{\text{sim}(z_i(n), z_j(n)^+)/\tau}}. \quad (19)$$

The above process details CAFF with our designed FFU module. The filtering operation is computationally lightweight, so our method does not introduce significant additional computational overhead. The number of trainable parameters, GPU memory usage, and training time of our model are nearly the same as those of the traditional contrastive learning framework. After training, we obtain higher-quality embeddings.

4. Experiments

4.1. Dataset and experiment setup

Our experiments follow the settings of unsupervised SimCSE and are implemented using PyTorch. We use 10^6 randomly sampled sentences from English Wikipedia, as provided by SimCSE, to train the embeddings. For evaluation, we employ SentEval ([Conneau & Kiela, 2018](#)) on seven semantic textual similarity (STS) tasks and seven transfer tasks. [Table 1](#) presents an overview of all datasets and tasks used in our experiments.

Furthermore, following ([Chuang et al., 2022](#)), we conduct a retrieval experiment. In line with prior work, we use all the data from the STSB dataset (training, test, and development sets) for our retrieval experiments. The dataset contains 8628 sentence pairs with human-annotated similarity scores. The distribution of similarity scores is shown in [Fig. 3](#). Among these pairs, 419 are strong positive pairs (similarity score = 5). For retrieval, we use one sentence of each such pair to retrieve the other from a pool of 17,255 sentences.

4.2. Training details

We use the SimCSE checkpoint as the initial encoder for our model. An MLP layer is first employed to transform the embeddings to $d_k = 1024$, after which FFT is used for frequency transformation. For the convolutional layer, we set $m = 16$ for the BERT model and $m = 18$ for the RoBERTa model. In the mask fusion phase, we choose $\alpha = 0.7$. The model is trained for one epoch. We use the Adam optimizer with a batch size of 64 and a temperature parameter $\tau = 0.05$ in the contrastive loss. The learning rate is set to 9×10^{-6} (9e-6). Training the embeddings on an Nvidia 3090 GPU takes approximately 3 h.

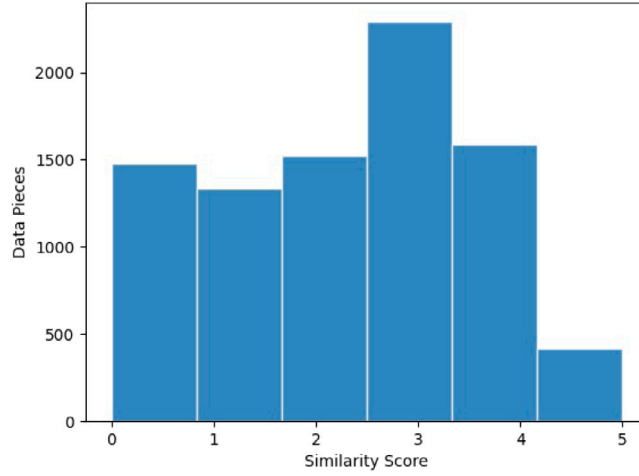


Fig. 3. The similarity score distribution of the STSB dataset.

Table 2

Sentence embedding performance on STS tasks measured by Spearman's correlation. $\diamond$ Results from Klein and Nabi (2023); $\dagger$ results from Xu et al. (2023); $\heartsuit$ results from Gao et al. (2021); $\S$ results from Chen et al. (2023); $\P$ results from Jeong et al. (2024). Our results are highlighted. Results are shown as mean $\pm$ standard deviation from multiple independent experiments (same hereinafter).

Model	STS12	STS13	STS14	STS15	STS16	STSB	SICKR	Avg
BERT_{base}								
BERT	21.54	32.11	21.28	37.89	44.24	20.29	42.42	31.40
BERT-flow $\diamond$	58.40	67.10	60.85	75.16	71.22	68.66	64.47	66.55
Consert $\dagger$	64.64	78.49	69.07	79.72	75.95	73.97	67.31	72.74
SimCSE $\heartsuit$	68.40	82.41	74.38	80.91	78.56	76.85	72.23	76.25
SSCL $\S$	71.68	83.50	76.42	83.46	78.39	79.03	71.76	77.90
SimCSE++ $\dagger$	73.66	82.36	75.86	83.09	79.76	79.71	71.91	78.05
miCSE $\diamond$	71.71	83.09	75.46	83.13	80.22	79.70	73.62	78.13
SimACE $\P$	71.63	83.44	76.65	83.85	79.95	79.99	71.86	78.20
CAFF	72.72 ± 0.21	83.31 ± 0.57	76.22 ± 0.21	83.64 ± 0.19	80.53 ± 0.28	80.59 ± 0.2	73.07 ± 0.32	78.58 ± 0.37
BERT_{large}								
BERT	27.74	30.76	22.59	29.98	42.74	26.75	43.44	32.00
SimCSE $\heartsuit$	70.88	84.16	76.43	84.50	79.76	79.26	73.88	78.41
CAFF	73.59 ± 1.17	85.70 ± 0.37	77.65 ± 0.19	85.28 ± 0.23	80.83 ± 0.85	81.34 ± 0.65	75.45 ± 0.26	79.98 ± 0.62
RoBERTa_{base}								
RoBERTa	17.45	41.36	26.98	52.15	54.97	38.82	61.90	41.95
SimCSE $\heartsuit$	70.16	81.77	73.24	81.36	80.65	80.22	68.56	76.57
CAFF	72.27 ± 0.18	82.55 ± 0.31	75.12 ± 0.17	82.85 ± 0.29	81.61 ± 0.28	81.25 ± 0.20	69.91 ± 0.62	77.93 ± 0.34
RoBERTa_{large}								
RoBERTa	21.49	25.62	13.74	36.90	39.66	17.30	40.63	27.91
SimCSE $\heartsuit$	72.86	83.99	75.62	84.77	81.80	81.98	73.88	78.90
CAFF	75.11 ± 0.27	84.64 ± 0.88	77.47 ± 0.15	85.4 ± 0.11	81.26 ± 0.42	82.81 ± 0.16	72.71 ± 0.93	79.91 ± 0.51

Table 3

Transfer task results measured by accuracy. All models are based on BERT_{base}.

Model	MR	CR	SUBJ	MPQA	SST	TREC	MRPC	Avg
BERT	78.68	84.85	94.21	88.23	84.13	91.40	71.13	84.66
SimCSE	81.18	86.46	94.45	88.88	85.50	89.80	74.43	85.87
CAFF	81.37 ± 0.63	85.96 ± 0.69	94.57 ± 0.21	89.15 ± 0.14	85.04 ± 0.19	89.58 ± 0.46	76.38 ± 0.43	86.01 ± 0.29

4.3. Baselines

Given that there are no existing studies on frequency domain filtering of embeddings, we select BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), BERT-flow (Li et al., 2020), SimCSE (Gao et al., 2021), and ConSERT (Yan et al., 2021) as our baselines. These models include both traditional encoders and contrastive learning-based encoders. In addition, we compare against current state-of-the-art contrastive learning models, including SimCSE++ (Xu et al., 2023), SSCL (Chen et al., 2023), miCSE (Klein & Nabi,

2023), and SimACE (Jeong et al., 2024). All models are non-large language models and are trained on the same Wiki1M dataset provided by SimCSE. Using consistent evaluation metrics – SentEval for STS tasks and cosine similarity-based top- k retrieval for retrieval tasks – we compare our model’s performance with these baseline methods.

4.4. Main results

4.4.1. STS tasks

Table 2 presents our STS results. For in-depth analysis and comparison, we focus on BERT_{base}-based models as representatives, while also providing a brief comparison across other models. Compared with the baselines, our method yields substantial improvements and achieves the best overall performance, demonstrating that even a simple frequency filtering mechanism can effectively reduce semantic redundancy and produce higher-quality embeddings. In particular, CAFF improves over SimCSE by 2.33% and 1.57% on BERT base and large models, and by 1.36% and 1.01% on RoBERTa base and large models, respectively, supporting our claim that filtering semantic redundancy enhances embedding quality. Furthermore, CAFF attains the highest metrics among all compared methods, highlighting its potential for general-purpose sentence representation.

The observed performance variations can be explained by the intrinsic properties of each dataset (Table 1). STS12 shows the largest gain (+4.32%) as a heterogeneous collection of multiple subtracks spanning diverse domains, where semantic filtering effectively mitigates domain-specific noise and enhances sensitivity to fine-grained distinctions. Similarly, STS15 (+2.73%) and STS-B (+3.74%) are large and linguistically diverse, making them highly responsive to embedding improvements. STS14 exhibits moderate heterogeneity, yielding noticeable yet smaller gains, whereas STS16 is relatively small and task-focused, constraining potential improvements. In contrast, STS13 (+0.90%) is small and domain-homogeneous, and SICK-R (+0.84%) consists of synthetically constructed, repetitive sentence pairs; in both cases, the limited linguistic diversity reduces sensitivity to embedding refinements.

These results also have a plausible cognitive interpretation. Although it is difficult to directly attribute improvements to genuine cognitive mechanisms rather than statistical pattern learning, our approach is inspired by findings in cognitive science suggesting that humans evaluate semantic similarity by filtering out redundant or frequent information to focus on novel, informative cues. The consistent gains observed across diverse datasets suggest that cognitive alignment may contribute to model robustness. In STS tasks where human raters assign similarity scores, CAFF’s improved ability to capture fine-grained semantic distinctions can be viewed as an indirect reflection of this cognitive principle.

Overall, CAFF demonstrates the greatest advantage on large, heterogeneous datasets, while improvements are naturally constrained on smaller or more homogeneous benchmarks, confirming that semantic filtering is particularly effective when fine-grained discrimination across diverse semantic relations is required.

4.4.2. Transfer tasks

In this task, a logistic regression classifier is trained on top of (frozen) sentence embeddings. As noted by Gao et al. (2021), transfer tasks are not the primary goal of sentence embeddings; therefore, we provide a simple comparison. Table 3 shows our results using BERT and SimCSE on the base model. On average, CAFF outperforms BERT by 1.35% and SimCSE by 0.14% across seven transfer benchmarks. We attribute these relatively modest improvements mainly to the misalignment between how the downstream classifiers are trained and the cognitive principle underlying our approach. Specifically, logistic regression operates directly on the frozen sentence embeddings and utilizes all frequency information for decision making, whereas CAFF is designed to filter out high-frequency redundant content. This mismatch in training objectives limits the classifier’s ability to fully exploit the benefits of semantic redundancy filtering. From cognitive perspective, transfer tasks – especially those framed as single-sentence or sentence-pair classification – still benefit from improved representation quality, but the standard training setup may not allow the classifiers to leverage the nuanced semantic mechanisms that humans naturally use. In other words, the limited gain is not due to the method being unsuitable, but rather to the task-specific training not fully aligning with the cognitively-inspired design of CAFF. Nonetheless, these results demonstrate that cognitively-inspired improvements in representation quality remain stable even when used as features in traditional classifiers, and motivate further research on classifier architectures and training strategies that better align with the principles of semantic redundancy filtering.

4.4.3. Retrieval task

Effective retrieval is fundamental to model performance, as it enables accurate and efficient access to relevant information. Here, we demonstrate the retrieval performance of our model. In the sentence retrieval task, we follow (Chuang et al., 2022) and evaluate the quality of sentence embeddings by retrieving the counterpart of each sentence based on cosine similarity. Specifically, for each sentence pair, one sentence is used as a query to retrieve its counterpart, and we determine whether the correct match appears among the top 1, 5, or 10 most similar sentences. As shown in Table 4, CAFF consistently achieves the highest recall@1, @5, and @10, surpassing both BERT and SimCSE. These results confirm that frequency-based filtering substantially enhances the distinctiveness of sentence representations, thereby enabling more accurate retrieval of relevant sentences. From cognitive perspective, this improvement resonates with the mechanisms by which humans disambiguate meaning: filtering out redundant cues and selectively attending to distinctive, task-relevant information. By explicitly reducing the influence of frequent components in the embedding space, CAFF mirrors the human ability to highlight semantic novelty and contextual relevance—both of which are fundamental to effective recall and matching in natural language understanding. The consistent gains across recall metrics therefore not only demonstrate the statistical effectiveness of our approach, but also provide evidence that cognitively inspired redundancy filtering can serve as a powerful enhancement for retrieval-oriented applications.

Table 4
Retrieval task results based on BERT_{base}.

Model/Recall	@1	@5	@10
BERT	28.64	37.23	40.09
SimCSE	76.13	95.22	97.85
CAFF	78.04	96.18	99.52

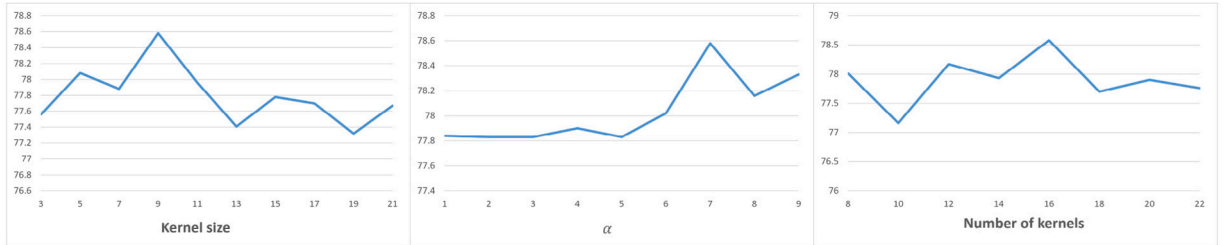


Fig. 4. Performance with different hyperparameter settings.

4.5. Ablation studies

4.5.1. Choice of hyper-parameters

We conducted a grid search to evaluate the impact of key hyperparameters: kernel size k , number of kernels m , and α . For each hyperparameter, we performed extensive experiments while keeping the other two fixed. The results, shown in Fig. 4, demonstrate that CAFF consistently achieves strong performance across a wide range of hyperparameter combinations. To further analyze parameter sensitivity, we quantified the variation in accuracy for each parameter group. Specifically, the accuracy fluctuates within a narrow range as the values of k , m , and α change: the maximum difference for kernel size is 1.26, for number of kernels is 1.42, and for α is 0.75. The standard deviation for all groups is below 0.45, indicating minimal sensitivity to hyperparameter choices. No sharp drops or spikes in performance are observed, highlighting the robustness of our method. Therefore, the limited performance variation across different choices of kernel size, number of kernels, and α confirms that our design is robust and does not rely on exhaustive hyperparameter tuning, which greatly facilitates practical deployment.

4.5.2. Skipping initial embeddings

Our method modulates the frequency spectrum of the initial embeddings through our designed FFU, which raises a question: are the initial embeddings truly necessary? To answer this, we conduct an ablation experiment. As shown in Table 5, our model's performance decreases by 1.27% without the initial embeddings. This result indicates that the initial embeddings are essential, as they provide a frequency spectrum that better aligns with human cognition compared to embeddings without pre-training. Therefore, our adaptive frequency filtering can select the most suitable frequencies for semantic understanding.

4.5.3. Effect of the two filters

To further explore the effectiveness of our two adaptive filter masks, we conduct ablation experiments using only one mask at a time. As shown in Table 5, removing the linear mask leads to a 0.56% drop in performance, while removing the convolutional mask results in a 0.75% decrease. These results demonstrate that both the convolutional filter, which provides local contrastive enhancement, and the linear filter, which offers global compensation, are necessary. The comparison also indicates that the convolutional filter is more important than the linear filter. Therefore, in filter fusion, we assign a larger coefficient to the convolutional filter.

4.5.4. Importance of FFD

After filtering, we introduce an FFD layer designed to capture richer and more complex relationships within the frequency domain. To further investigate the impact of the FFD layer, we conduct an ablation study in which the FFD layer is removed. As shown in Table 5, without the FFD layer, our model's performance decreases by 1.63%. This result highlights the importance of the FFD layer in modeling interactions in the frequency domain.

5. Analysis

5.1. Different frequency transform

In this section, we conduct additional experiments using three different frequency transformations. As shown in Table 6, DFT achieves the highest performance among the three transformations. Each transformation represents a different approach for decomposing a vector into frequency components: DFT decomposes the vector into separate sine and cosine bases, DHT uses the

Table 5Results of CAFF ablations without first-stage training, convolutional filter, or linear filter based on BERT_{base}.

CAFF	Avg.STS
-all	78.58
-w/o initial embeddings	77.31
-w/o convolution filter	77.83
-w/o linear filter	78.02
-w/o FFD	76.95

Table 6Results of CAFF with different frequency transformations. Statistical significance is confirmed by ANOVA ($p = 6.642 \times 10^{-13}$).

Frequency transform	Avg.STS
DFT	78.58 _{± 0.37}
DCT	77.91 _{± 0.29}
DHT	77.74 _{± 0.16}

sum of sine and cosine bases, and DCT relies solely on the cosine basis. Although the differences in average performance and standard deviation among DFT, DCT, and DHT appear small, we performed a one-way analysis of variance (ANOVA) to determine whether these differences are statistically significant. The resulting p -value is 6.642×10^{-13} , which indicates that the differences among the three methods are highly significant and unlikely to be due to chance. Therefore, the conclusion regarding DFT's superior performance is statistically supported. Overall, DFT provides the most thorough and general decomposition, which explains why the DFT-based frequency filtering unit yields the best results. We conclude that a more comprehensive and general decomposition of the vector leads to significantly better performance.

5.2. Frequency spectrum and case study

We take a closer look at the frequency spectrum in NLP and give a case study. We first introduce how we obtain the embedding-level spectrum and the dataset-level spectrum, and then present our case study. Specifically, we calculate the embedding-level spectrum and subsequently aggregate it to obtain the dataset-level spectrum.

5.2.1. Embedding-level spectrum

We first use the encoder to generate the sentence embedding $x(n) = [x(1), x(2), \dots, x(N)] \in \mathbb{R}^N$, and then transform it into a frequency domain vector $f(u) = [f(1), f(2), \dots, f(N)]$ of the same dimension N via frequency transformation. For the frequency domain vector f , the complex value at each dimension represents the frequency component at that frequency. We then calculate the modulus of each frequency value to obtain the frequency spectrum: $spec(u) = [|f(1)|, |f(2)|, \dots, |f(N)|]$.

5.2.2. Dataset-level spectrum

For each sentence s_i in the dataset, we calculate the embedding spectrum $spec_i$ of the corresponding sentence embedding x_i using the method described in Section 5.2.1. We then sum the frequency spectra of all sentences to obtain the dataset-level spectrum: $spec_{data} = \sum_{i=1}^n spec_i$. The dataset-level frequency spectrum characterizes the overall frequency distribution of the dataset.

5.2.3. Case study

Fig. 5 presents a test case comparing SimCSE and CAFF, along with the visualized frequency spectra of two different sentences. In this example, CAFF achieves a cosine similarity score that is 5.1 points higher than SimCSE. We attribute this improvement to the trained frequency spectrum; CAFF exhibits a more discriminative spectrum compared to SimCSE. By leveraging this enhanced frequency spectrum, CAFF is able to provide more accurate semantic information, resulting in a higher similarity score.

Beyond the simple case study, we further calculate the frequency spectrum of the embeddings generated from the entire STSB dataset to assess universality. The results are shown in Fig. 6. We observe that the spectrum of CAFF covers the entire frequency range, with increased emphasis on certain frequencies. This selective focus suppresses semantic redundancy and provides higher-quality embeddings. Importantly, such emphasis is not merely a byproduct of statistical learning. Rather, it is guided by the design of the CAFF training mechanism, which encourages embeddings to form frequency patterns that align with human selective attention. The consistency of these patterns across the dataset demonstrates that the observed alignment is driven by cognitive plausibility rather than mere statistical coincidence. Notably, this behavior differs from the CV domain, where suppressing high frequencies typically leads to better feature representations. We next analyze the impact of high-frequency components.

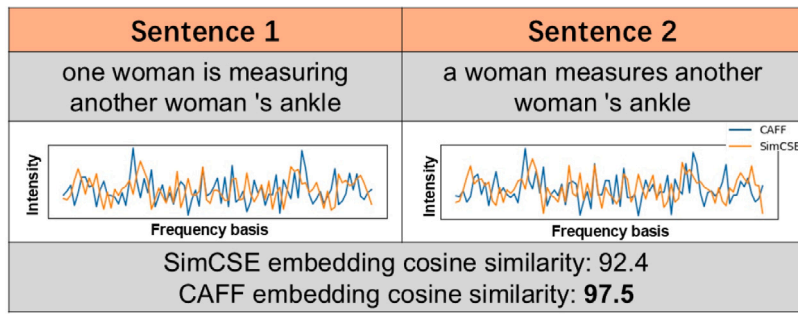


Fig. 5. Case study from the STSB dataset.

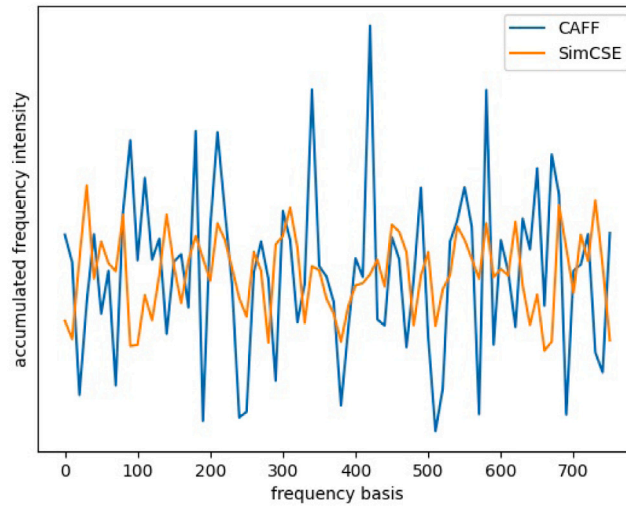


Fig. 6. The frequency spectrum of STSB dataset.

Table 7

Results of CAFF with different percentages of frequency components removed.

Frequency retention rate	Avg.STS
100%	78.58
90%	77.95
80%	78.01
70%	77.89
60%	77.80

5.3. Filtering high frequency off?

In the field of CV, humans perceive high-frequency information poorly, so filtering out high-frequency components is typically beneficial. By analogy, it may be reasonable to hypothesize that, in natural language processing (NLP), high frequencies could also be a source of discrepancy between human cognition and machine learning. To examine the effect of high-frequency components, we conduct an experiment in which we progressively cut off the high-frequency part of the spectrum by different percentages. Table 7 shows the results under various retention rates. The results indicate that removing a portion of high-frequency components leads to a decrease in model performance on the STS task. This finding suggests that high-frequency information is important for machine learning in NLP, which is different from the pattern observed in CV, and highlights the need to explore alternative filtering strategies.

6. Theoretical and practical implications

Our paper offers critical theoretical insights into the cognitive alignment of machine-learned representations in NLP. First, we introduce the novel concept of frequency redundancy as a fundamental cognitive gap between neural network-based sentence

embeddings and human semantic processing. This gap is empirically demonstrated, as random frequency filtering leads to substantial improvements in semantic similarity scores, with STSB performance increasing from 74.1 to 89.4. Second, the proposed CAFF method shows that lightweight frequency filtering via the Frequency Filtering Unit (FFU) can significantly enhance embedding quality through unsupervised training, without the need for costly architectural modifications. This approach enables easy generalization across downstream tasks and reduces the need for extensive data annotation. Third, the success of CAFF in NLP not only provides a new perspective for improving embedding quality but also suggests new possibilities for advancing downstream tasks.

In summary, our work not only proposes a new theoretical understanding of semantic representation from cognitive perspective but also offers a scalable and efficient solution to enhance embedding quality, with broad applicability in NLP and beyond.

7. Conclusion and future work

In this paper, we analyze the potential cognitive gap between current sentence embeddings and semantic information in the frequency domain, drawing inspiration from human cognition. This cognitive gap is manifested as semantic redundancy. To bridge this gap, we propose the CAFF training method, which employs an adaptive Frequency Filtering Unit (FFU) to modulate frequency components in accordance with human selective attention. The FFU extracts and removes non-selected frequencies from the initial embeddings, thereby reducing the impact of semantic redundancy and producing higher-quality embeddings, as verified by our experiments.

For future work, we will further examine the cognitive impact of high-frequency vectors on machine learning models and explore how insights from cognitive science can be systematically integrated into NLP, aiming to extend the applicability of our approach to broader tasks. We believe this direction offers a promising bridge between cognitive science and machine learning.

8. Limitation

A key limitation of this study is the absence of experimental validation directly linking CAFF's filtering mechanism to human cognitive processes. While selective attention in language perception serves as the theoretical foundation for our approach – suggesting that humans inherently prioritize certain frequencies during language processing – we currently lack direct experimental evidence to confirm that the prioritization patterns identified by CAFF correspond to human sensitivity to linguistic frequencies.

To address this limitation more rigorously, future research could involve developing behavioral experiments or analyzing psycholinguistic datasets that directly investigate the correspondence between CAFF's filtering patterns and human frequency sensitivity. Although such experiments have not yet been conducted, studies analogous to established research on human frequency perception in language processing, such as those by Ellis (2002) and Christiansen and Chater (2016), would be especially valuable. These efforts would further ground the CAFF approach in cognitive science principles.

CRedit authorship contribution statement

Chenrui Mao: Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Conceptualization. **Kai Shuang:** Writing – review & editing, Supervision. **Jinyu Guo:** Writing – review & editing, Supervision. **Bing Qian:** Supervision. **Yu Yang:** Writing – review & editing, Supervision. **Haoqing Li:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

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